

Ethan Dunne

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Professional Experience

Alarm.com - iOS Software Engineer I & II

Tysons, VA | August 2021 - Present

- CarPlay Integration: Created an Apple CarPlay driving task app for controlling a unit's garage doors as well as running scenes to control other devices in the home.
- System Control Widget: Created a widget on iOS on an expedited release timeline using SwiftUI, WidgetKit, and AppIntents for controlling your home security system without needing to open the main app.
- Saved Clips Widget: Created a widget on iOS for the user to view their latest video clips from their security cameras. Developed microApi and caching system to reduce database calls.
- Dashboard Load Time Improvements: Converted app home page from one large dashboard to a collection of cards that each make separate concurrent requests.
- App Diagnostics: Created a system for collecting logs from the app remotely for debugging purposes.

Skills employed: **Swift | SwiftUI | Objective-C | UIKit | WidgetKit | Combine | C# | .NET | SQL/SSMS | Python | Git | Atlassian Cloud**

Ekotrope - Software Engineer Intern

Boston, MA | May - August 2020

- Worked on a team developing for a java web application for rating homes (HERS ratings and other standards). Back-end uses specific algorithms to determine if buildings are certified.

Skills employed: **Java | Google Web Toolkit | CSS | Mercurial | MySQL**

Northbridge Companies - Software Intern

Burlington, MA | May - August 2019

- Developed a native iOS application; a central resource hub for a farm stand project that hosts events and sells merchandise.
- App users can obtain information about upcoming events, buy merchandise, and even order food remotely.
- Integrated an API called "molten" to create, keep stock of, and retrieve products from online as a placeholder backend for the merchandise store feature.

Skills employed: **Swift | UIKit**

Personal Projects

DC Vault Pole Vault App

October 2023 - Present

- Built a cross-platform app for my pole vault club using React Native.
- The app consists of many features including athlete profiles, keeping track of jumps, ranking athletes, displaying trends, unit conversions, scheduling classes, and more!
- Expanded upon their website's Node.js backend and MySQL database.

Skills employed: **Typescript | React Native | Expo Router | Node.js | MySQL | App Store Connect | Google Play Console | Full Stack Deployment**

Battle Blobs iOS Game

January - March 2023

- Built a 2D iOS game with singleplayer and local multiplayer.
- Implemented an upgrade system with multiple skills and balanced rewards for defeating enemies.
- Developed system for increasing enemy AI difficulty based on level number.
- Published on the Apple App Store.

Skills employed: **Swift | SwiftUI | SpriteKit | App Store Connect | Game Design**

Runic Commander Computer Game

- Building a top-down 2D game in Unity
- The game features many complex systems and controls, such as an inventory system, crafting system, building system, skills/experience system, and many more.
- Designed AI for non-player characters using finite state machines.
- Created all art assets and animations.

Skills employed: **C# | Unity3D | Aseprite | Game Design**

Sound-Reactive LED Lights

June 2018 - 2020

- Wired up LED strips controlled by an Arduino Uno and transistors.
- Use a microphone to capture audio levels and convert it to colors with custom sound level to color map.
- github.com/edunne4/Apartment-Lighting

Skills employed: **C++ | Microcontrollers | Electronics and Circuit Design | Soldering | Networking**

Education

Bucknell University

Lewisburg, PA | August 2017 - May 2021

- Bachelor of Science in Computer Engineering, **3.93 GPA**
- Honors:
 - Summa Cum Laude | Bucknell Prize in Computer Engineering | 2018-19 Google Cloud Academic All-District Team (CoSIDA) | 2020 Patriot League Indoor Track and Field Academic All-League Team | Dean's List | Track and Field Varsity Captain
- Jobs:
 - Software Engineering and Design Course - Teaching Assistant | Teaching and Learning Center - Math and Physics Tutor
- Relevant Courses:
 - Operating Systems Design | Design & Analysis of Algorithms | Human-Computer Interaction | Machine Learning | Computer Organization | Individual Study on Object Detection and Vision for Autonomous Robots | Image Processing | Software Engineering & Design | Probabilistic System & Data Analysis | Data Structures & Algorithms | Digital System Design | Signals & Systems Theory | Embedded & Cyber-Physical Systems